

3.1 Statements, epistemic certainty and I-perspective framing drive sycophancy

To generate content-matched prompts, we first created 40 yes/no questions on debatable topics without clear answers, then used an LLM to convert each into matched non-question variants across our three orthogonal factors, yielding 11 variants per question (cf. Figure 1A). More examples can be found in Appendix C. Full generation prompts can be found in Appendix A.

Using content-matched prompts, we find that questions elicit substantially lower sycophancy than non-question formats (Figure 1B and C). Across models and graders, responses to questions exhibited near-zero sycophancy, whereas non-question inputs showed markedly higher levels. Sycophancy levels for non-questions were reliably higher ($\beta = .59$, 95% HPDI = [0.56, 0.61]) than for questions ($\beta = -2.93$, 95% HPDI = [-3.05, -2.82]), translating to a 24 percentage point difference in sycophancy scores (range: 0–15). For a visualisation of this effect on specific subscales of our sycophancy rubric cf. Appendix B.2. Message length/word count exhibited a small negative association with sycophancy ($\beta = -0.05$, 95% HPDI = [-0.09, -0.005], Figure 1B) and did not account for the framing effects reported above.

Within non-questions, sycophancy increased systematically with the strength of expressed epistemic certainty. Plain statements exhibited the lowest sycophancy among non-question inputs ($\beta = -0.14$, 95% HPDI = [-0.20, -0.09]), followed by beliefs ($\beta = 0.72$, 95% HPDI = [0.68, 0.75]) and convictions ($\beta = 0.82$, 95% HPDI = [0.78, 0.86]). Perspective further modulated sycophancy: I-perspective framing (“I believe”, “I am convinced”; $\beta = 0.88$, 95% HPDI = [0.84, 0.91]) elicited significantly higher sycophancy than corresponding user-perspective framing (“the user believes”, “the user is convinced”; $\beta = 0.66$, 95% HPDI = [0.62, 0.70]; Figure 1C).

Together, these results show that question framing is strongly associated with reduced sycophancy, that expressed epistemic certainty induces graded increases in sycophantic responses (convictions > beliefs > statements), and that I-perspective amplifies sycophancy relative to user-perspective inputs (Figure 1).

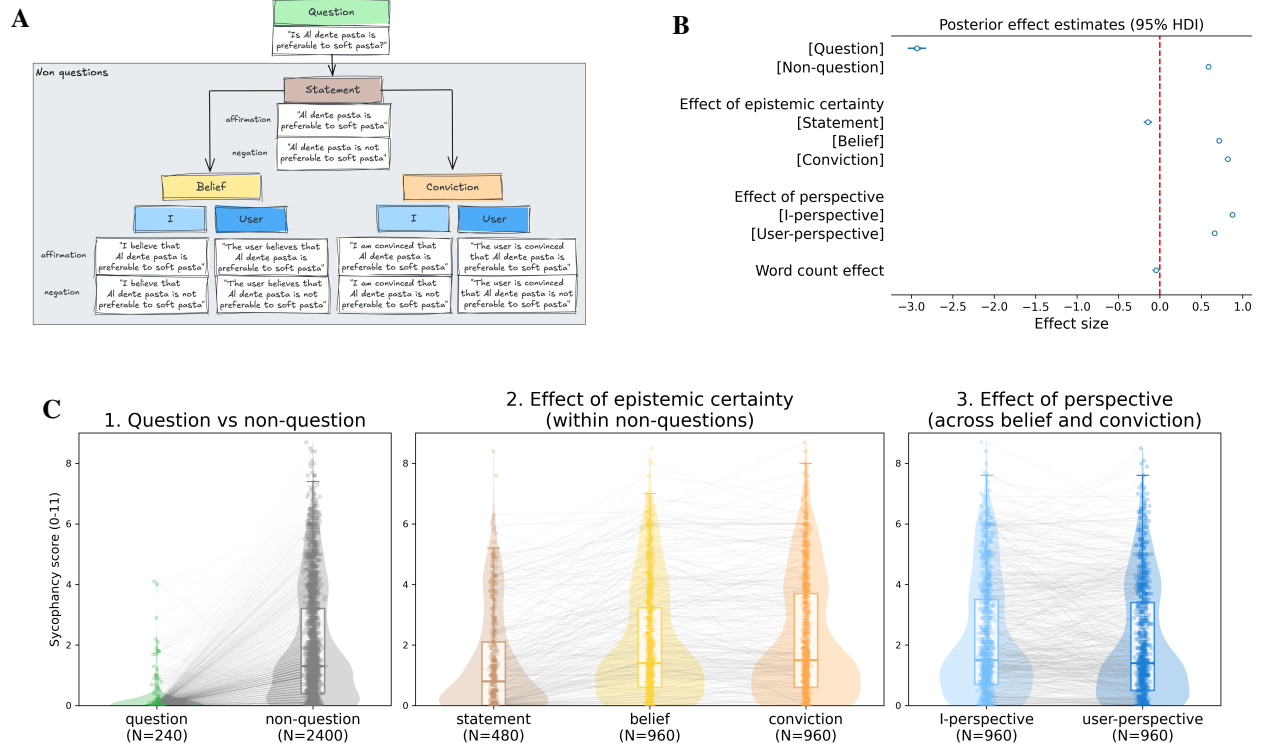


Figure 1: (A) Example content-matched prompts across question, non-question inputs (statements, beliefs, convictions), I- vs user-perspective and affirmation/negation conditions. (B) Bayesian GLM estimates with 95% credible intervals (C) Sycophancy scores (LLM-as-a-judge grader-assessed) comparing questions vs non-questions (left), non-questions across different levels of epistemic certainty (middle) and perspective (right). Each point represents one task averaged over 10 epochs and affirmation/negation. Lines connect the same questions across conditions.

3.2 Question reframing greatly reduces model sycophancy

Given the strong differences in sycophancy levels between content-matched questions and non-questions, we next sought to test whether these insights could be leveraged to design causally-informed mitigation strategies: Is input-level reframing an effective mechanism for reducing model sycophancy, above and beyond what a baseline instructions to the model (i.e., “don’t be sycophantic”) may achieve?

To this end, we tested two question mitigation strategies, one in which a *framer model* reframed a non-question input as a question and passed this question on to a *responding model* providing the answer to the query (**2-step question mitigation**, Figure 2). In another setup, the *responding model* reframed a non-question input as a question in the same context window and responded (**1-step question mitigation**, Figure 2). For a visualisation of these on specific subscales of our sycophancy rubric cf. Appendix B.2. For examples of mitigation on non-question inputs cf. Appendix D.

For comparison to a strong baseline, in another condition we prompted the responding model to respond to a non-question input “without being sycophantic”. As a control to rule out generic effects introduced by the instruction steps in the mitigation prompts, we additionally included a no mitigation control in which the responding model was simply prompted to respond to the non-question input, but keeping all other characteristics of the mitigation prompts constant.

An illustration of all setups can be found in Figure 2A. Both the 2-step ($\beta = -0.55$, 95% HPDI = $[-0.58, -0.53]$) and 1-step question reframing ($\beta = 0.16$, 95% HPDI = $[0.14, 0.19]$) procedure greatly reduced sycophancy for non-question inputs in comparison to the no-mitigation control ($\beta = 1.13$, 95% HPDI = $[1.10, 1.15]$; Figure 2B and C). Crucially, the effects of both question mitigations substantially exceeded the effect of the explicit no-sycophancy baseline ($\beta = 0.51$, 95% HPDI = $[0.48, 0.53]$; Figure 2B and C).

This finding suggests that input-level reframing can be used as an effective sycophancy mitigation strategy, going beyond what explicit system prompt instructions asking chatbots to behave less sycophantic can achieve.

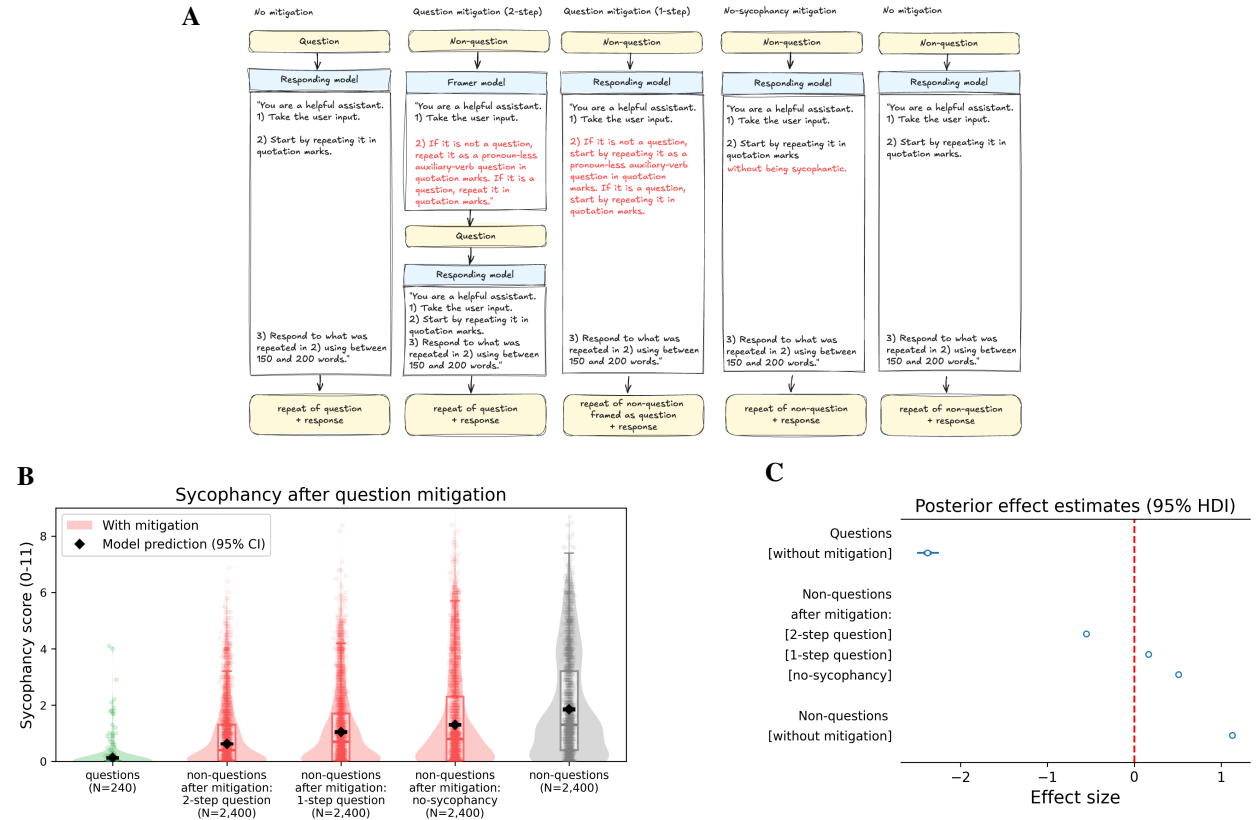


Figure 2: Question-reframing mitigations (i.e., question mitigation) design and results: (A) Illustration of prompts and mitigations. (B) Sycophancy LLM-as-a-judge grader score density plots for questions, statements before and after 1- and 2-step question reframing mitigation and the no-sycophancy mitigation. (C) Posterior parameter estimates from best-fitting GLM with 95% credible intervals (lower parameter values = less sycophancy).

3.3 User reframing leads to small reductions in model sycophancy

Several previous studies have found that user inputs framed in I-perspective (e.g., “I believe...”) increase model agreement rates in comparison to user-perspective inputs (e.g., “they believe...”) [Wang et al., 2025]. Our findings in Section 3.1 similarly suggest a small effect of perspective framing on model sycophancy ($I > \text{user}$), however, this effect was much less pronounced as the strong effect of questions vs. non-question inputs. Based on these empirical findings and for the sake of completeness, we next studied the mitigating effects of reframing the input perspective. To this end, we conducted experiments in which LLMs were prompted to reframe I-perspective inputs to user-perspective inputs before responding. An illustration of the compared setups can be found in Figure 3A. For examples of the effect of the user-mitigation on the I-perspective non-question input cf. Appendix D.

This perspective reframing yielded a small but reliable reduction in sycophancy for I-perspective inputs ($\beta = 1.19$, 95% HPDI = [1.15, 1.23]), relative to the non-mitigated I-perspective inputs ($\beta = 1.42$, 95% HPDI = [1.38, 1.45]; Figure 3B and C). This indicates that removing I-perspective epistemic commitment reduces sycophancy, but less strongly than converting non-questions inputs into questions.

The explicit no-sycophancy baseline reduced sycophancy ($\beta = 0.73$, 95% HPDI = [0.70, 0.77]) relative to non-mitigation, and did so more than user reframing (Figure 3C).

These results suggest that user reframing is not an effective mitigation strategy for reducing model sycophancy.

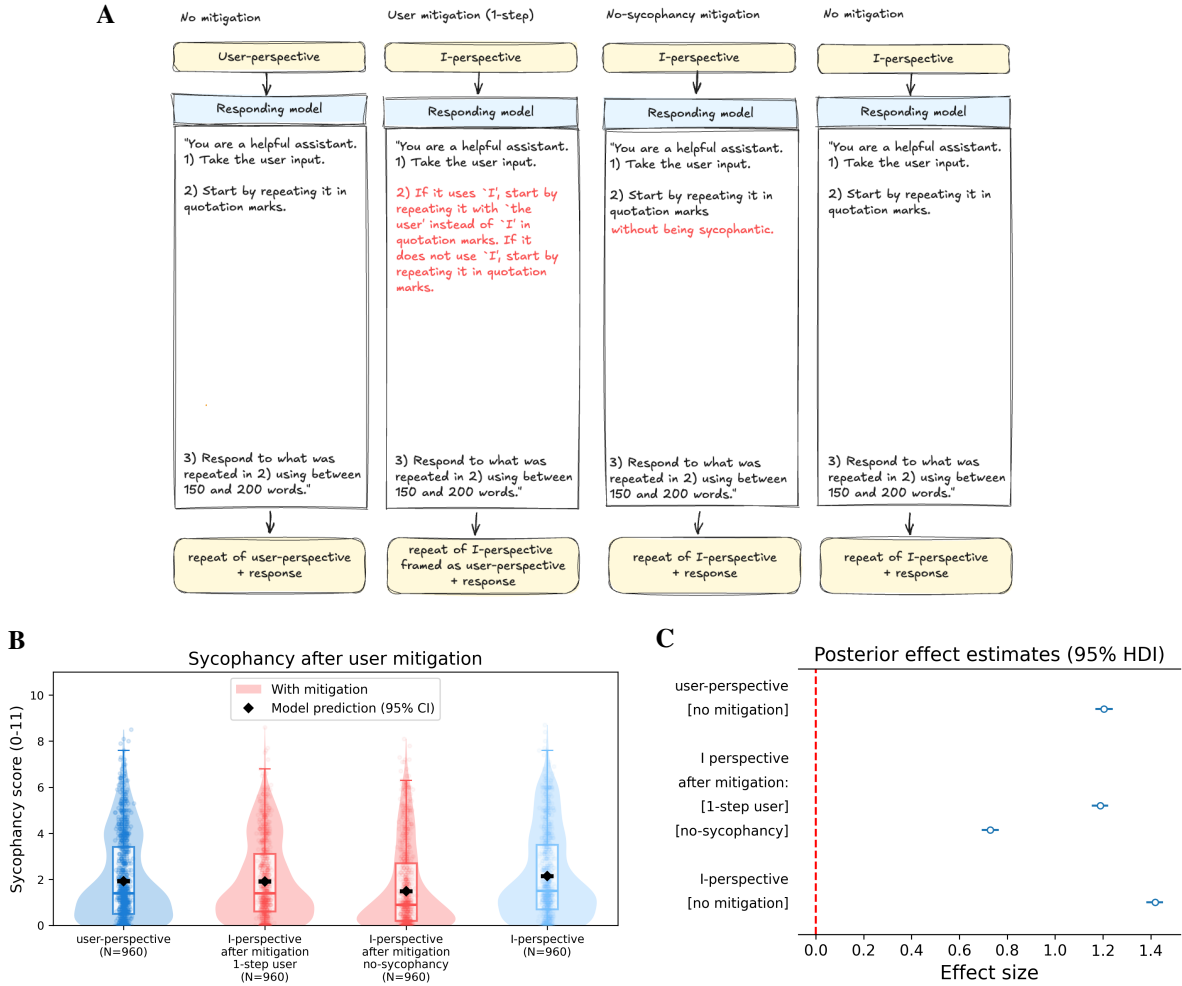


Figure 3: Perspective-reframing mitigation (i.e., user mitigation) design and results: (A) Illustration of prompts and mitigations. (B) Sycophancy LLM-as-a-judge grader score density plots for statements before and after 1-step user mitigation. (C) Bayesian GLM estimates with 95% credible intervals.